

Occlusion-Adaptive Temporal Memory: A Camera-Only Framework for Track Continuity Under Visual Occlusion in Autonomous Driving

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Abstract

Camera-based perception for autonomous driving relies almost entirely on single-frame object detectors, which have no inherent mechanism for distinguishing a temporarily occluded object from one that has genuinely left the scene. This work presents Occlusion-Adaptive Temporal Memory (OATM), a causal, camera-only tracking method that extends ByteTrack’s two-stage, confidence-aware association with explicit target-occluder relational reasoning: when a mature track goes unmatched, OATM evaluates whether a currently visible object plausibly explains the disappearance, stores the target in an occluder-centric coordinate frame, checks that memory against the target’s own independent motion prediction, predicts when the occlusion should end, and restricts identity reconnection to a bounded, relation-aware association stage. In a final scene-disjoint evaluation on ten nuScenes-mini scenes, with five used only to fix lifecycle thresholds and five held out for validation, every method ran causally on identical frozen detections without ground-truth input. Only after the tracker outputs were saved did a LiDAR-supported offline 3D evaluation project the official annotations into the camera image and score 1,873 car and pedestrian annotations at IoU 0.30. OATM reached 84.1% precision, 40.4% F1, 18.4% MOTA, and 30.3% IDF1, with 59 identity switches, 28 fragmentations, and 20.055 px mean center error, versus 73.4% precision, 40.1% F1, 14.2% MOTA, and 29.6% IDF1 for a five-frame ByteTrack buffer, and 55.3% precision, 37.7% F1, 1.1% MOTA, and 27.4% IDF1 for a twelve-frame buffer. OATM also produced the most reliable hidden-state predictions (39.4% predicted-hidden precision versus 25.0% and 14.0%) and the fewest unsupported keyframe tracks (7.8% versus 25.9% and 22.1%), evidence of selective relational persistence rather than an indiscriminately longer wait. These results support a stronger balanced accuracy-identity-persistence tradeoff than either tested ByteTrack buffer, though OATM’s overall recall (26.6%) and its recall in the most-occluded visibility bin (8.3%) remained slightly below both baselines, and the evaluation is a ten-scene nuScenes-mini study rather than a full benchmark.

1. Introduction

Robust object tracking under occlusion remains a fundamental challenge for camera-based perception in autonomous driving. Single-frame detectors estimate location and class independently for each image and have no mechanism for reasoning about continuity across time, so a temporarily occluded object and one that has genuinely left the scene both manifest identically as an absence of detection. On reappearance, the object is typically registered as a new, unrelated detection, severing identity continuity, a real cost for downstream planning and control, which depend on continuous position estimates of nearby road users. Occlusion is not a rare failure mode: in the nuScenes dataset [1], nearly 40% of annotated object instances exhibit less than 80% visibility at some point in their track.

Tracking-by-detection methods associate per-frame detections into tracks using a motion model, typically a Kalman filter under a constant-velocity assumption, combined with an intersection-over-union (IoU)-based assignment solved via the Hun-

garian algorithm [2]. Subsequent work extends association to incorporate low-confidence detections that would otherwise be discarded [3], and separate lines of research aggregate visual features temporally [4] or maintain longer-horizon memory of tracked objects [5–8], reviewed in Section 2. Existing methods, however, generally treat occlusion as an implicit byproduct of detector failure or a fixed confidence threshold rather than a distinct state requiring explicit classification, and none jointly provides a memory of the hidden target explicitly protected against contamination by the occluder, a termination rule tied to a predicted end of occlusion, and a reconnection stage guarded against identity hijacking by nearby same-class objects.

Without an explicit, causally reasoned notion of occlusion, a tracker cannot decide whether to keep predicting a hidden object’s position, terminate it as lost, or protect its remembered state from being silently overwritten by whatever is occluding it. To address this gap, we propose Occlusion-Adaptive Temporal Memory (OATM), a causal, camera-only tracking method built on the same two-stage, confidence-aware association and constant-velocity motion prediction as ByteTrack [3], extended with three components. First, when a mature track goes unmatched, an explicit target-occluder relation forms only if a currently visible object plausibly explains the disappearance, moving through a defined lifecycle (forming, active, clearing, resolved, or failed) rather than being treated as an undifferentiated association failure. Second, the hidden target is stored in a coordinate frame centered on its occluder and accepted only while it agrees with the target’s own independently predicted position, so an inconsistent or drifting relation is rejected rather than followed. Third, OATM predicts the expected moment the occlusion should end and couples this to a bounded reappearance stage and a priority-ordered termination rule, so a track is retired for exactly one traceable reason rather than a fixed frame-count timer. The resulting contribution is fourfold: an explicit and auditable target-occluder lifecycle; an occluder-centric geometric memory guarded by an independent Kalman prediction; a clearance-aware, spatially bounded same-identity reconnection stage; and a scene-disjoint evaluation that keeps causal camera tracking strictly separate from LiDAR-supported 3D annotation scoring. The final comparison also separates observed outputs from predicted-hidden outputs, making persistence reliability measurable rather than hiding temporal predictions inside an aggregate track count. Sections 3 and 4 describe this method and its evaluation in full.

2. Related Works

Recent progress in multi-object tracking has been driven substantially by improvements in object detection. Modern detectors, including the YOLO family [9], estimate bounding boxes and class labels independently for each frame; this per-frame independence is exactly what prevents a detector with no access to prior frames from distinguishing a temporarily occluded object from one that is genuinely absent, and the tracking literature has largely developed to compensate for this rather than to remove it.

The dominant paradigm, tracking-by-detection, treats detections as a fixed observation stream and associates them into coherent identities separately. SORT [2] does so with a Kalman filter under a constant-velocity assumption and an IoU cost matrix solved by the Hungarian algorithm, but associates only detections above a fixed confidence threshold; a detection below it, whether from occlusion, blur, or scale, is discarded and becomes indistinguishable from a genuine absence of evidence. ByteTrack [3] addresses this with a two-stage procedure: high-confidence detections matched first, then a second round against low-confidence detections, never allowing an unmatched low-confidence detection to start a new track. This recovers weak

visual evidence, but once a detection is entirely absent, ByteTrack has no mechanism beyond generic motion extrapolation until its track buffer expires.

A related line of work addresses degraded detection quality at the feature level. Flow-guided feature aggregation [4] aligns convolutional features across neighboring frames using estimated optical flow, improving consistency for objects still partially visible nearby, an assumption that fails under multi-frame full occlusion. MeMOT [5] and MeMOTR [6] use transformer-based memory to retain object representations across longer sequences than a Kalman filter alone tolerates. PermaTrack [7] trains an explicit object-permanence objective, predicting continued existence through full occlusion rather than relying on motion extrapolation. Mask-guided spatio-temporal feature aggregation [8] restricts temporal aggregation using segmentation masks, reducing the risk that an occluding object contaminates the representation of the target it hides.

Occlusion may also be addressed with sensing unavailable to a single camera: LiDAR and radar can, depending on occluder geometry, provide evidence invisible to the camera [10], and cooperative perception shares observations across vehicles or infrastructure to cover a single vehicle’s blind spots [11]. These directions are complementary to the present work, which stays camera-only to characterize what temporal reasoning alone can recover.

Across this body of work, three limitations recur: occlusion is handled as an incidental consequence of a detection or association failure rather than an explicitly classified perceptual state; temporal memory of a hidden target, where retained at all, is not explicitly protected against contamination or hijacking by the object doing the occluding, often of the same class; and no reviewed method couples memory retention to a predicted end of occlusion together with a principled, bounded reconnection rule, relying instead on a fixed frame-count threshold, an implicit confidence cutoff, or a learned model’s opaque memory. OATM addresses these three limitations directly, as described in Section 3.

3. Methodology

3.1 Problem Formulation and Design Requirements

Let I_t be the current CAM_FRONT image, D_t the detector’s set of class-labeled boxes and confidence scores, and S_{t-1} the complete tracker state accumulated before frame t . The online tracker is a causal mapping

$$(Y_t, S_t) = \mathcal{F}(I_t, D_t, S_{t-1}), \quad (1)$$

where Y_t contains the frame’s saved track outputs. Equation 1 excludes future images, future detections, and all official annotation or sensor metadata. Each output is marked either as an observed box supported by a current detection or as a predicted-hidden box supported only by the track’s temporal state; a temporal prediction is never presented as a current visual detection.

For a mature target that becomes unmatched, the decision is not simply how long to retain the track. OATM must determine whether a visible object currently explains the disappearance, whether that explanation remains geometrically consistent, when separation should occur, and whether a later detection can safely inherit the original identity. These requirements lead to four design constraints: persistence must be conditional on present occlusion evidence; relational geometry must be checked

against an independent motion prediction; reappearance must use bounded one-to-one association; and every termination must have a traceable cause. The online architecture in Figure 1 summarizes this decision path.

3.2 Dataset and Detection

All experiments use the publicly released mini subset of nuScenes [1], restricted to the forward-facing camera channel (CAM_FRONT) only, so the study isolates what temporal reasoning alone can recover from a single monocular stream. The mini subset comprises 10 scenes and 2,342 total CAM_FRONT frames, of which 404 are human-annotated keyframes (2 Hz); the remainder are unannotated "sweep" frames. Ground truth is constructed independently of the tracking pipeline: nuScenes' 3D bounding-box annotations for the car and pedestrian categories are projected into the CAM_FRONT image plane using recorded ego-vehicle pose and camera calibration [1], yielding 3,492 in-scope projected car and pedestrian annotations across all 404 keyframes. The ten scenes are split whole, never at the frame level, into five development scenes, used only to fix lifecycle thresholds, and five held-out validation scenes, whose 1,873 annotations are the sole basis for every comparison reported in Section 4. Projected annotations, visibility labels, calibration, LiDAR/radar information, and recorded ego pose are used exclusively to prepare and evaluate events offline; none is ever an input to the online tracker.

A single pretrained, lightweight YOLO detector [9] is run once, in prediction-only mode at a confidence floor of 0.05, over every frame in all 10 scenes on CPU, producing 49,436 fresh detections in 860.4 seconds. Detections are cached so every compared tracking method receives an identical, frozen detection stream; only the car and pedestrian categories are used downstream. Processing every frame before consulting any annotation preserves the causal, online setting used by tracking-by-detection systems such as SORT and ByteTrack [2, 3].

3.3 OATM: Occlusion-Adaptive Temporal Memory

OATM retains ByteTrack's two association rounds [3] unchanged: high-confidence detections matched to predicted tracks, then remaining tracks given one further opportunity against low-confidence detections, and activates its own reasoning only once these ordinary rounds fail to observe a mature target. At that point OATM asks a question ByteTrack itself does not ask: is this disappearance actually explained by a visible occluder, and when should that explanation stop being trusted?

3.3.1 Occlusion Candidate Scoring and Relational Lifecycle

For an unmatched, mature target box T and visible candidate box O , target-relative coverage is

$$C(T, O) = \frac{\text{area}(T \cap O)}{\text{area}(T)}. \quad (2)$$

OATM combines this coverage with occluder-to-target scale, a depth-order cue from the boxes' lower image boundaries, trajectory agreement, track maturity, and optional camera-motion quality. Writing these normalized features as $\phi_k(T, O)$, the candidate score is

$$q(T, O) = \sum_k w_k \phi_k(T, O), \quad O^* = \arg \max_O q(T, O), \quad (3)$$

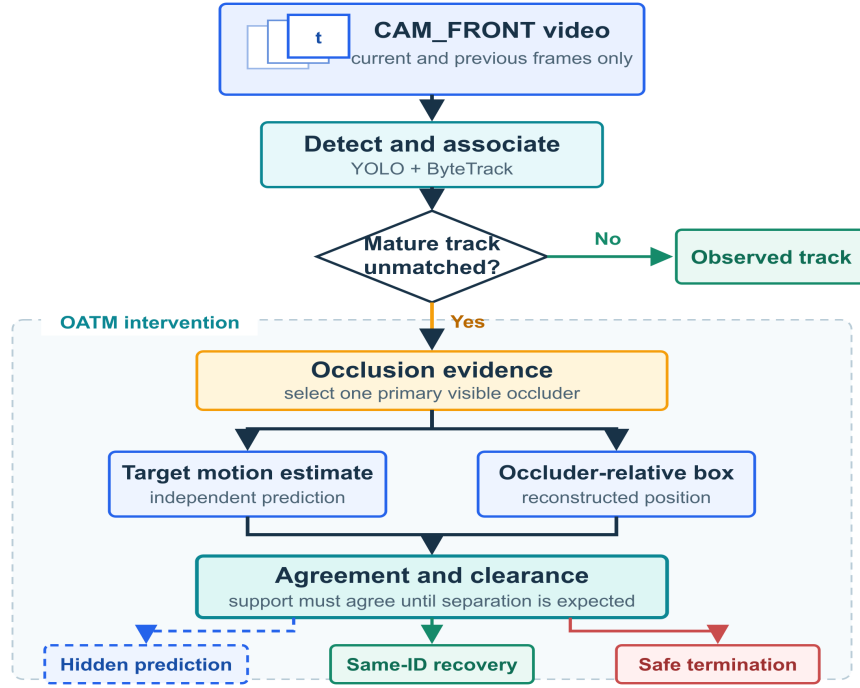


Figure 1. OATM online workflow used in the project presentation. Ordinary YOLO–ByteTrack association produces observed tracks; only a mature, unmatched target enters the OATM branch. Occlusion evidence then combines independent target motion with occluder-relative reconstruction to support a labeled hidden prediction, restore the same identity, or terminate the track safely.

and a relation may form only when $q(T, O^*) \geq 0.48$ and the coverage and scale gates also pass. The weighted score in Equation 3 is a deterministic engineering signal, not a calibrated probability. A supported target forms exactly one primary relation with its highest-scoring occluder, moving through a defined lifecycle: FORMING, when a visible occluder first explains the disappearance; ACTIVE, while the relation stays geometrically consistent (Section 3.3.2); CLEARING, once separation is predicted or support becomes inconsistent; RESOLVED, when a valid detection reconnects the original identity; and FAILED, when persistence ends without recovery. The primary occluder cannot switch mid-episode, so a different nearby object can never silently take over ownership of the target.

3.3.2 Occluder-Centric Geometric Memory

Rather than remembering only a hidden target’s last appearance, OATM stores its geometry relative to the primary occluder. For target center and size (c_x^T, c_y^T, w_T, h_T) and occluder geometry (c_x^O, c_y^O, w_O, h_O) at relation formation, the stored state is

$$\delta_x = \frac{c_x^T - c_x^O}{w_O}, \quad \delta_y = \frac{c_y^T - c_y^O}{h_O}, \quad r_w = \frac{w_T}{w_O}, \quad r_h = \frac{h_T}{h_O}. \quad (4)$$

At a later frame, the visible occluder decodes the implied target center as $(c_x^O + \delta_x w_O, c_y^O + \delta_y h_O)$ and its size as $(r_w w_O, r_h h_O)$. The decoded anchor is accepted only when its center lies within

$$\rho_{\max} = \max\left(20 \text{ px}, 0.75\sqrt{\widehat{w}_T^2 + \widehat{h}_T^2}\right) \quad (5)$$

of the target’s independent causal Kalman prediction, and each decoded scale ratio is between $1/1.25$ and 1.25 . If either check fails, OATM keeps the independent prediction and moves the relation to CLEARING instead of allowing the occluder-relative anchor to drag the target. Figure 2 illustrates this dual-prediction guard.

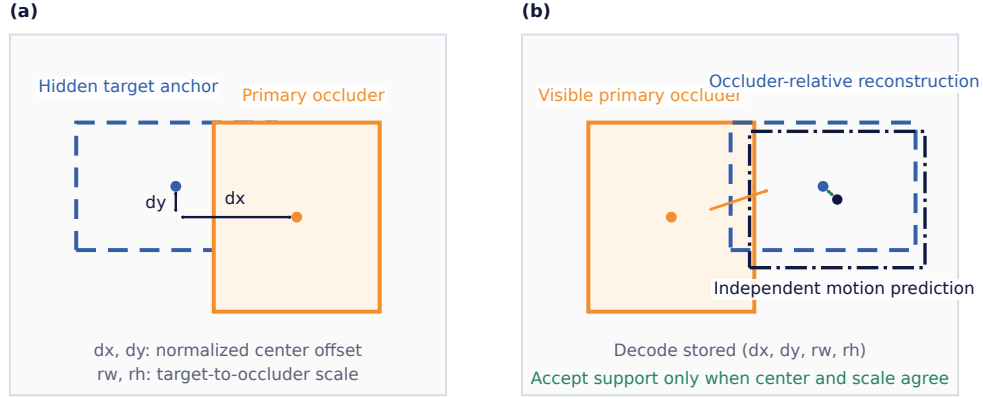


Figure 2. Occluder-relative geometric memory. OATM stores the target’s normalized offset and scale relative to one primary visible occluder. On later frames, the reconstructed box is trusted only when it agrees with the target’s independent motion prediction in center and scale.

3.3.3 Expected Clearance, Termination, and Protected Reappearance

While a relation is active, OATM propagates both boxes over a bounded twelve-frame horizon. The expected clearance is the earliest step

$$h^* = \min\{h \in \{1, \dots, 12\} : C(\hat{T}_{t+h}, \hat{O}_{t+h}) < 0.05\}, \quad (6)$$

when such a step exists. After h^* , the target receives a four-frame reappearance grace window rather than indefinite persistence. A mature target that never forms an occlusion relation at all is not dropped immediately either: after its ordinary one-frame output grace expires, it may hold its identity for one further silent frame, emitting no box at all, and reconnects only through a separate, higher-threshold check; failing that check ends the identity. A hidden target is terminated the first time any one condition holds: outward motion toward an image boundary with less than 60% of the predicted box still visible, an immature track, no occlusion relation beyond the ordinary one-frame grace and its single silent frame, expected clearance passing without reappearance, hidden duration exceeding twelve frames, or localization uncertainty exceeding a configured bound. Persistence is therefore always tied to one specific, traceable reason.

A relationally hidden target is excluded from ByteTrack’s own association rounds; reconnection instead uses a separate, stricter, one-to-one stage scoring remaining detections on class agreement, distance to the independently predicted box, scale agreement, and consistency with the expected clearance time, accepting a match only above 0.55 and within an uncertainty-expanded search radius hard-capped at 150 px. An active occluder blocks reconnection unless the relation has already entered CLEARING. This prevents a nearby same-class object from hijacking the target’s identity, and prevents a distant, unrelated weak detection from being accepted merely because the ordinary buffer has not yet expired. A mature unsupported track that never forms a relation receives one additional silent DORMANT frame, emits no box, and must pass the stricter 0.75

reappearance threshold to restore its identity.

The resulting lifecycle is causal: a matched observation can become a relation-supported hidden prediction only after a visible occluder explains its disappearance. OATM then propagates explicitly labeled temporal predictions until expected clearance, after which it either reconnects a valid observation to the original identity or terminates unsupported persistence. Every saved output row distinguishes these predicted boxes from current visual detections.

A camera-motion compensation branch (background feature tracking and robust translation estimation) was implemented and evaluated as an ablation but is disabled in the promoted configuration and was not used for the final reported experiment, because it reduced error on a controlled panning scenario while producing severe drift on longer natural sequences once combined with the image-space velocity already modeled by the target’s own Kalman filter.

3.4 Auditable Outputs and Final Parameterization

OATM preserves the provenance of every reported track box. As summarized in Table 1, current visual evidence and temporal prediction are distinct states, and the relational diagnostics needed to explain persistence are saved beside each prediction. This prevents a hidden estimate from being mistaken for a detector observation during analysis.

Table 1. Saved output semantics. Relational fields are populated only when an occlusion relation exists.

Field or state	Meaning
OBSERVED_STRONG	Current high-confidence visual detection.
OBSERVED_WEAK	Current low-confidence visual detection recovered by the second association round.
PREDICTED_HIDDEN	Temporal estimate with no current detector support; never described as a detection.
evidence_source	Strong detection, weak detection, or motion prediction.
occluder_track_id	Immutable primary occluder for the current hidden episode.
relation_diagnostics	Candidate score, lifecycle phase, expected-clearance horizon, and localization uncertainty.

The final thresholds in Table 2 were selected on the five development scenes before the validation scenes were scored. They expose the method’s operating scale and make clear that the twelve-frame duration is an upper bound rather than an unconditional retention time.

Table 2. Promoted OATM configuration used for the final scene-disjoint experiment.

Component	Value	Component	Value
Detection floor	0.05	High-score threshold	0.50
New-track threshold	0.60	Relation score threshold	0.48
Target coverage gate	0.12	Clearance coverage	0.05
Maximum hidden duration	12 frames	Reappearance grace	4 frames
Relation reappearance	0.55	Dormant reappearance	0.75
Search-radius cap	150 px	Minimum residual allowance	20 px
Maximum scale ratio	1.25	Exit visible fraction	0.60
Silent dormant window	1 frame	Camera compensation	disabled

3.5 Evaluation Protocol and Metrics

OATM is compared against ByteTrack itself at two track-buffer lengths, using identical cached detections and identical association thresholds [3], so OATM’s relational reasoning is the single controlled variable. ByteTrack-5 uses a five-frame buffer, matching this project’s earlier baselines; ByteTrack-12 uses a twelve-frame buffer matching OATM’s own maximum hidden-duration ceiling, so any OATM advantage cannot be attributed to a longer generic wait than its baseline is given.

3.5.1 Online Causal Evaluation Stage

In the online stage, the frozen detector and each tracker process all 2,342 CAM_FRONT frames strictly in timestamp order. At frame t , a method may use only the current image, the current detections, the current timestamp, and its state from frames up to t ; future frames and official annotations are unavailable. The tracker saves observed detections and temporally predicted hidden states as distinct output types. This stage therefore measures the behavior of a causal camera-only system rather than a tracker using privileged sensor or annotation information, consistent with the online protocols described for SORT and ByteTrack [2, 3].

3.5.2 Offline LiDAR-Supported 3D Evaluation Stage

Only after every tracker output has been saved does a separate offline evaluator load the official nuScenes 3D boxes, instance identities, visibility tokens, sensor calibration, ego pose, and LiDAR/radar point counts [1]. The 3D boxes are projected into CAM_FRONT, and tracker outputs are matched to those projected 2D boxes with class-aware, one-to-one Hungarian assignment. Thus, “LiDAR-supported 3D evaluation” refers to the provenance and metadata of the official annotations used for retrospective scoring; LiDAR points, 3D boxes, visibility, calibration, and ego pose are never inputs to the online detector, ByteTrack baselines, or OATM. Figure 3 shows this one-way evaluation boundary.



Figure 3. Strictly separated LiDAR-supported 3D evaluation. Frozen camera-only tracker outputs are combined with projected nuScenes annotations, visibility labels, and LiDAR/radar support metadata only after online tracking has completed. Class-aware Hungarian matching produces the reported metrics, with no feedback to the tracker.

The final comparison uses all ten nuScenes-mini scenes, split whole into five development scenes, used only to fix lifecycle and consistency thresholds, and five held-out validation scenes, used only for the reported comparison in Section 4. Tracker output and official 3D annotations are matched after tracking finishes, offline, using class-aware one-to-one Hungarian matching at an IoU of 0.30, with IoU 0.10 and 0.50 checked as sensitivity gates [2]; the 1,873 validation-scene annotations are the sole basis for every reported number.

Table 3 separates the online processing population from the annotated offline scoring population. Unannotated sweep

frames are processed causally because they affect tracker state, but they are never treated as empty-ground-truth frames.

Table 3. Dataset and evaluation populations. Scene assignments are whole; neighboring frames never cross the development/validation split.

Population	Count
nuScenes-mini scenes	10
Development / validation scenes	5 / 5
Causal CAM_FRONT frames processed online	2,342
Official annotated keyframes scored offline	404
Projected car/pedestrian annotations, all scenes	3,492
Validation annotations for final comparison	1,873
Frozen detector outputs supplied to every method	49,436

nuScenes visibility tokens (four coarse bins from roughly 0-40% to 80-100% annotated-visible fraction) and LiDAR/radar point counts accompany each official annotation [1] and are used only offline, as sensitivity strata for stratifying recall by occlusion severity and sensor support, never as an online occlusion signal and never to remove hard examples from the reported denominator. A zero-LiDAR-point annotation is not treated as occluded, since an object can register zero points from distance or sparse sampling alone; it is retained in the headline population like every other annotation. This coarse, image-level label is not an exact per-pixel occlusion mask, so precise occlusion start, end, and duration still require manual CAM_FRONT review.

Table 4. nuScenes visibility strata used only for offline sensitivity analysis. The bins describe coarse annotated visibility, not exact per-camera masks.

Token	Annotated visible fraction	Interpretation in this study
1	0-40%	Most-occluded visibility bin
2	40-60%	Heavily obscured
3	60-80%	Partly obscured
4	80-100%	Most-visible visibility bin

Reported metrics include precision, recall, and F1 against the matched annotations, plus identity switches and fragmentation counts. Following CLEAR MOT and the identity measures of Ristani et al. [12, 13], the two aggregate tracking scores are

$$\text{MOTA} = 1 - \frac{\sum_t (\text{FN}_t + \text{FP}_t + \text{IDSW}_t)}{\sum_t \text{GT}_t}, \quad (7)$$

$$\text{IDF1} = \frac{2 \text{IDTP}}{2 \text{IDTP} + \text{IDFP} + \text{IDFN}}. \quad (8)$$

MOTA in Equation 7 combines misses, false positives, and identity switches, whereas IDF1 in Equation 8 emphasizes identity-consistent detections. The evaluation also reports mean center error between matched predicted and annotated box centers; predicted-hidden precision, the fraction of temporally predicted outputs that match an annotation; and an unsupported-keyframe-track rate, a sparse-label proxy for ghost persistence measuring how often a tracker output at an annotated keyframe corresponds to no official annotation. Every threshold used in Section 4 was fixed on the five development scenes before the five validation scenes were scored, and the run carries a single identifier (lidar-fixes-20260805) so every reported number is

traceable to its source.

4. Results and Discussion

The headline evidence in this section comes only from the scene-disjoint validation population of run lidar-fixes-20260805. All three methods use the same frozen camera detections and the same 1,873 validation annotations at the primary IoU 0.30 gate. The qualitative case study in Section 4.3 explains one identity lifecycle but is not used to enlarge the final validation claim.

4.1 Final Scene-Disjoint Evaluation

On the five held-out validation scenes (1,873 annotations, IoU 0.30), OATM reached 84.1% precision, 40.4% F1, 18.4% MOTA, and 30.3% IDF1, with 59 identity switches, 28 fragmentations, and 20.055 px mean center error. ByteTrack-5 reached 73.4% precision, 40.1% F1, 14.2% MOTA, and 29.6% IDF1, with 64 identity switches, 33 fragmentations, and 21.084 px error. ByteTrack-12 reached 55.3% precision, 37.7% F1, 1.1% MOTA, and 27.4% IDF1, with 82 identity switches, 33 fragmentations, and 21.361 px error.

Table 5 consolidates the complete comparison. The upper panel reports detection and identity scores; the lower panel isolates localization and persistence reliability so predicted-hidden behavior is not obscured by aggregate metrics.

Table 5. Final scene-disjoint comparison on 1,873 validation annotations at IoU 0.30. All methods use identical frozen camera detections. Bold values mark the best result in each column.

Method	Prec.	Recall	F1	MOTA	IDF1	IDSW	Frag.
ByteTrack-5	73.4	27.6	40.1	14.2	29.6	64	33
ByteTrack-12	55.3	28.6	37.7	1.1	27.4	82	33
OATM	84.1	26.6	40.4	18.4	30.3	59	28

Method	Center error (px)	Pred.-hidden prec.	Unsupported rate	False pred.-hidden
ByteTrack-5	21.084	25.0%	25.9%	150
ByteTrack-12	21.361	14.0%	22.1%	393
OATM	20.055	39.4%	7.8%	40

Figure 4 makes the principal tradeoff visible: OATM leads precision, F1, MOTA, and IDF1, while both fixed-buffer baselines retain slightly more objects overall.

4.2 Persistence Reliability and Identity Quality

OATM also reached 39.4% predicted-hidden precision and a 7.8% unsupported-keyframe-track rate, versus 25.0% and 25.9% for ByteTrack-5 and 14.0% and 22.1% for ByteTrack-12; in absolute counts, OATM produced only 40 false predicted-hidden matches against 150 for ByteTrack-5 and 393 for ByteTrack-12.

Figures 5 and 6 show that the benefit is not confined to one aggregate score. OATM led both baselines on every metric above except recall. F1, MOTA, and IDF1 together indicate the strongest balanced detection-and-identity result; the fewest identity switches and fragmentations indicate the most stable identity assignment; the lowest center error indicates the best localization; and the combination of the highest predicted-hidden precision with the lowest unsupported-track rate indicates that OATM’s persistence is selective rather than merely longer. ByteTrack-12’s own numbers make the contrast explicit:

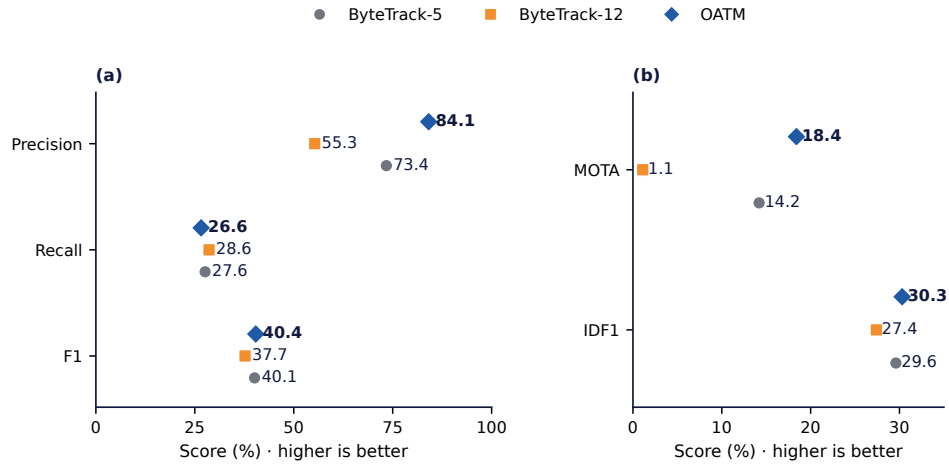


Figure 4. Final scene-disjoint validation profile. OATM achieved the highest precision, F1, MOTA, and IDF1 among the matched methods, while both ByteTrack settings retained slightly higher recall. Values are percentages.

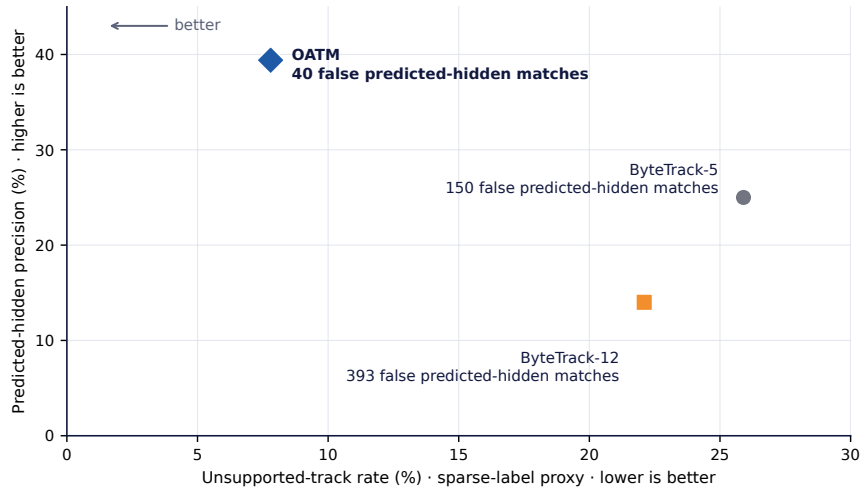


Figure 5. Selective persistence tradeoff. OATM combines the highest predicted-hidden precision with the lowest unsupported-track rate and only 40 false predicted-hidden matches. Unsupported-track rate is a sparse-annotation proxy, not a fully verified ghost-duration measure.

giving it OATM’s own maximum hidden duration did not translate into a better balanced result, since its identity switches, fragmentations, precision, and unsupported-track rate were all worse than OATM’s, and its MOTA fell even below the shorter ByteTrack-5 buffer’s. A longer generic wait is not, by itself, evidence toward the tradeoff OATM targets.

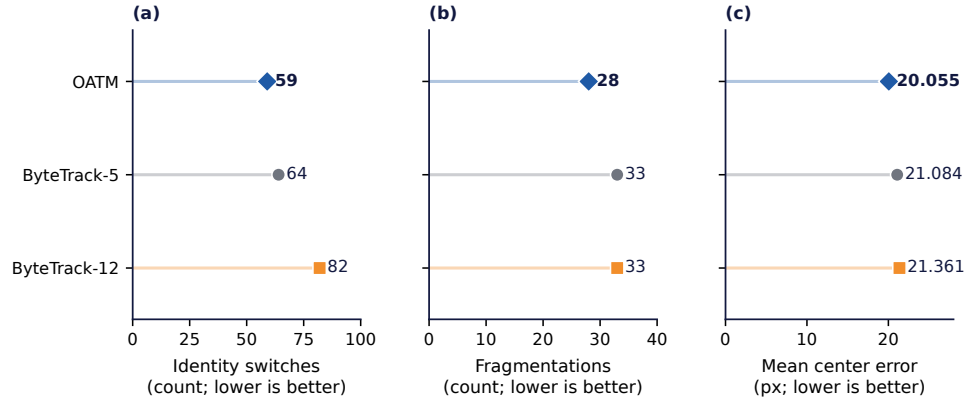


Figure 6. Identity continuity and localization on the final validation split. OATM produced the fewest identity switches and fragmentations and the lowest mean center error. Lower values are better in all three panels.

4.3 Qualitative Identity-Recovery Case Study

Aggregate metrics describe the frequency of identity errors but not the temporal behavior that produces them. Figure 7 therefore presents five chronological frames from the real-event case study used in the project poster. Before occlusion, both methods localize the target. Across the complete event, ByteTrack-5 changes the target identity from 45 to 57, 86, and finally 89; the intermediate ID 57 occurs between the displayed frames. OATM retains ID 47 before, during, and after the occlusion. The sequence is consistent with OATM’s lower aggregate identity-switch and fragmentation counts, while remaining an illustration of one event rather than an additional statistical estimate.



Figure 7. Qualitative identity continuity through one real occlusion event. Blue denotes OATM and orange denotes ByteTrack-5. OATM preserves target ID 47 across the displayed interval, whereas ByteTrack-5 has already changed from ID 45 to ID 86 by mid-occlusion and finishes with ID 89. The full event contains an additional intermediate ByteTrack-5 assignment, ID 57, not shown among these five selected frames.

4.4 Limitations

The gains are not uniform across metrics. OATM’s overall recall was 26.6%, slightly below ByteTrack-5’s 27.6% and ByteTrack-12’s 28.6%, and in the most-occluded visibility bin (annotated visible fraction under 40%) OATM’s recall fell to 8.3%, versus 9.3% and 10.7% for the two baselines, the clearest remaining target for the relation-formation mechanism specifically. The evaluation itself is a ten-scene nuScenes-mini study with 1,873 validation annotations, not a full-nuScenes benchmark or a statistically powered universal comparison, and its occlusion evidence is the coarse visibility label described in Section 3.5, not an exact per-camera pixel mask or a verified occlusion start and end time; the unsupported-track rate is likewise a sparse-label proxy rather than a directly verified ghost duration. This limitation follows in part from the dataset’s annotated-keyframe schedule [1]. All three methods share the same frozen detector, so objects too small, distant, or heavily obscured to produce any usable detection are outside what association (relational or otherwise) can recover.

The qualitative sequence in Figure 7 makes one identity lifecycle directly inspectable, but a single selected event cannot estimate an event-level success rate. Its role is explanatory: it connects the aggregate identity counts to a concrete failure mode without enlarging the validation population or the numerical claims.

Figure 8 makes the principal weakness explicit. Recall falls sharply for every method in visibility token 1, and OATM’s

8.3% is below both ByteTrack settings. This suggests that future work should improve relation formation for heavily obscured targets rather than merely relaxing the termination safeguards that produced the precision and false-persistence gains.

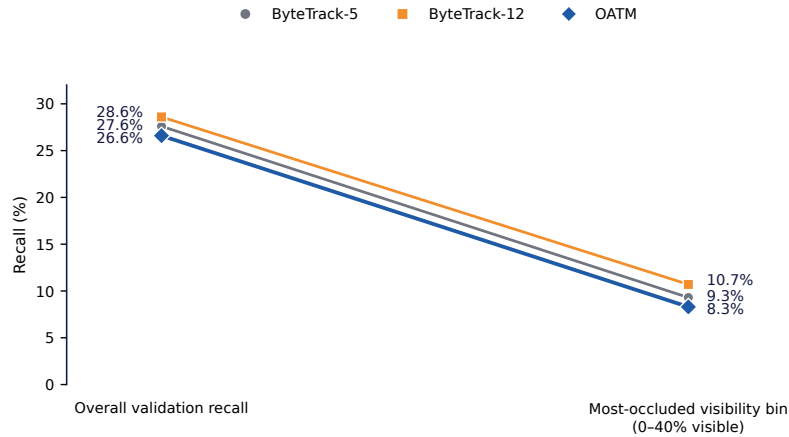


Figure 8. Recall under severe visibility. For nuScenes visibility token 1 (0–40% annotated visibility), OATM reaches 8.3% recall versus 9.3% for ByteTrack-5 and 10.7% for ByteTrack-12. The token is a coarse annotation bin, not an exact CAM_FRONT occlusion mask.

Taken together, these results support a specific, bounded set of claims. OATM is causal and camera-only at inference time; it distinguishes current detections from temporal predictions in every saved output; it achieved the strongest F1, MOTA, IDF1, identity-switch count, fragmentation count, localization error, predicted-hidden precision, and unsupported-track rate among the three evaluated methods on the final validation split. They do not support broader claims: OATM does not lead in recall, either overall or in the most-occluded bin; the occlusion-candidate score is a deterministic engineering signal, not a calibrated probability; and no result here reproduces a published MOT17, MOT20, or full nuScenes benchmark number, since this ten-scene mini comparison is specific to this project.

5. Conclusions

This work presented OATM, a causal, camera-only tracking method that extends ByteTrack with explicit target-occluder relational reasoning: occlusion-candidate scoring and lifecycle, occluder-centric geometric memory checked against independent motion prediction, expected-clearance termination with a single silent, high-threshold identity frame for a mature miss that never forms a relation, and protected reappearance association. In a final scene-disjoint evaluation on ten nuScenes-mini scenes, five held out for validation, OATM reached the strongest precision, F1, MOTA, IDF1, identity-switch count, fragmentation count, localization error, predicted-hidden precision, and unsupported-track rate among the tested methods, using identical frozen detections and identical association thresholds throughout.

Three conclusions follow. First, conditioning persistence on an explicit, causally scored occlusion relation, rather than an undifferentiated miss or a fixed buffer, is implementable from camera signals alone and produces a measurably stronger balanced accuracy-identity-persistence tradeoff than either tested ByteTrack buffer on this evaluation. Second, that advantage is not simply a longer wait: ByteTrack-12 was given OATM’s own maximum hidden duration and still trailed OATM on identity, localization, and false-persistence measures, and trailed even the shorter ByteTrack-5 buffer on MOTA. Selectivity,

not duration, is what the relational mechanism contributes. Third, the result is bounded rather than universal: OATM’s overall and most-occluded-bin recall remained slightly below both baselines, and the ten-scene, 1,873-annotation evaluation is a reproducible nuScenes-mini study, not a claim of universal ByteTrack superiority or a reproduced published benchmark.

The clearest next step is closing the recall gap in the most-occluded visibility bin, since this is where the relation-formation mechanism specifically underperforms rather than where its safeguards are too conservative. A full nuScenes evaluation, rather than the ten-scene mini subset used here, would support confidence intervals and per-class results instead of a single matched comparison. The single qualitative event in Figure 7 should therefore be extended to a manually verified event-level cohort. Such a cohort, rather than the coarse visibility-label stratification used here, would let identity recovery, ghost duration, and occlusion timing be checked directly instead of through the sparse unsupported-track proxy. Camera-motion compensation, evaluated as an ablation but disabled in the promoted configuration, should be reconsidered only once target state and ego-motion are fused within one stabilized coordinate system.

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7. References

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